

Introduction to NGINX

A Professional Web Server for Real Applications

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- We've used PHP's built-in server (`php -S`) for learning and testing.
- It's convenient but limited and not for production.
- Real apps use proper servers like **NGINX** (fast, efficient) or **Apache** (widely used).
- We use **NGINX** for ASE 230; Simpler, faster, and better for learning modern PHP deployment.

What is NGINX?

NGINX (pronounced "Engine-X") is a powerful, high-performance web server used by:

- **Netflix** - Video streaming
- **Airbnb** - Travel platform
- **GitHub** - Code hosting
- **Slack** - Team communication
- **Dropbox** - File storage

Key Features

- **High Performance:** Handles thousands of connections
- **Reverse Proxy:** Routes requests to backend applications
- **Static Files:** Serves images, CSS, JS efficiently
- **Load Balancing:** Distributes traffic across servers

NGINX vs PHP Built-in Server

Feature	PHP -S	NGINX
Purpose	Development only	Production ready
Performance	Limited	High performance
Concurrent Users	Few	Thousands
Static Files	Basic	Optimized
Configuration	None	Highly configurable
SSL/HTTPS	Not supported	Full support
Real World Usage	Never	Everywhere

Goal: Install NGINX and learn professional web server setup!

Installation NGINX

1. **Windows** - Using manual download (recommended for novices) or choco (recommended for advanced users)
2. **macOS** - Using Homebrew (easiest method)
3. **Linux** - Using package managers (apt/yum)
4. **Verification** - Confirming installation works

Prerequisites

- Administrator/sudo access
- Basic command line knowledge
- No existing web servers on port 80 (stop Apache if running)

Windows Installation

Method 1: Manual Download (Recommended for Novices)

Step 1: Download NGINX

1. Go to <http://nginx.org/en/download.html>
2. Download **nginx/Windows** (stable version)
3. Extract to `C:\nginx` (create folder if needed)

Step 2: Start NGINX (Three options)

Option 1 (Simplest with two Terminals)

Open two Terminals.

Start from one terminal (skip `cd C:\nginx` if in the directory).

```
cd C:\nginx  
nginx.exe
```

To stop, run this command from the 2nd terminal.

```
cd C:\nginx  
# stop  
nginx -s quit
```

Option 2 (One Terminal)

Open a Terminal.

```
cd C:\nginx  
C:\nginx>start nginx.exe -p C:\nginx -c conf\nginx.conf  
# stop  
C:\nginx>nginx.exe -p C:\nginx -s quit
```

Option 3 (PowerShell)

Open PowerShell Terminal.

```
cd C:\nginx  
# stop  
PS C:\nginx> Start-Process -FilePath ".\nginx.exe" -ArgumentList '-p "C:\nginx" -c "conf\nginx.conf"  
PS C:\nginx> .\nginx.exe -p "C:\nginx" -s quit
```

Useful commands

```
taskkill /F /IM nginx.exe
```

Step 3: Verify Installation

You can check if the nginx is running.

```
> tasklist | findstr /I nginx
```

nginx.exe	79696	RDP-Tcp#0	1	9,492 K
nginx.exe	25920	RDP-Tcp#0	1	9,964 K

You can check if configuration is OK.

```
C:\nginx>.\nginx.exe -p "C:\nginx" -c "conf\nginx.conf" -t
nginx: the configuration file C:\nginx/conf\nginx.conf syntax is ok
nginx: configuration file C:\nginx/conf\nginx.conf test is successful
```

Open browser and go to: <http://localhost>

You should see: "Welcome to nginx!"

Method 2: Installing NGINX using Choco (Recommended for Advanced Users)

- Use the "4. Installing NGINX using Choco (Optional).md" for the installation guide.
- You need to understand choco, sc, and nssm to use this feature comfortably.

macOS Installation

Using Homebrew (Easiest Method)

Step 1: Install Homebrew (if not installed)

Open **Terminal** and run:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```


Step 2: Install NGINX

```
brew install nginx
```

Step 3: Start NGINX

```
# Start NGINX  
brew services start nginx  
  
# Or start manually  
nginx
```

Step 4: Verify Installation

Open browser and go to: <http://localhost:8080>

You should see: "Welcome to nginx!"

Linux Installation

Ubuntu/Debian (using apt)

Step 1: Update Package List

```
sudo apt update
```

Step 2: Install NGINX

```
sudo apt install nginx
```

Step 3: Start and Enable NGINX

```
# Start NGINX service
sudo systemctl start nginx

# Enable auto-start on boot
sudo systemctl enable nginx

# Check status
sudo systemctl status nginx
```

Verification Steps

1. Check if NGINX is Running

Windows:

```
tasklist | findstr nginx
```

macOS/Linux:

```
ps aux | grep nginx
```

2. Test Web Server

Open browser and navigate to:

- **Windows/Linux:** <http://localhost>
- **macOS:** <http://localhost:8080>

3. Check NGINX Version

```
nginx -v
```

You should see output like: `nginx version: nginx/1.18.0`

Basic NGINX Commands

Starting and Stopping NGINX

Windows:

```
# Start  
nginx.exe
```

```
# Stop  
nginx.exe -s stop
```

```
# Reload configuration  
nginx.exe -s reload
```

macOS:

```
# Start  
brew services start nginx  
# or: nginx  
  
# Stop  
brew services stop nginx  
# or: nginx -s stop  
  
# Reload  
nginx -s reload
```

Linux:

```
# Start
sudo systemctl start nginx

# Stop
sudo systemctl stop nginx

# Restart
sudo systemctl restart nginx

# Reload configuration
sudo systemctl reload nginx
# or: sudo nginx -s reload

# Check status
sudo systemctl status nginx
```


Important File Locations

Configuration Files

Windows (Chocolatey):

- Main config: `C:\tools\nginx\conf\nginx.conf`
- HTML files: `C:\tools\nginx\html\`

Windows (Manual):

- Main config: `C:\nginx\conf\nginx.conf`
- HTML files: `C:\nginx\html\`

macOS (Homebrew):

- Main config: `/usr/local/etc/nginx/nginx.conf`
- HTML files: `/usr/local/var/www/`

For Apple Silicon Mac:

- Main config: `/opt/homebrew/etc/nginx/nginx.conf`
- HTML files: `/opt/homebrew/var/www`

Linux:

- Main config: `/etc/nginx/nginx.conf`
- HTML files: `/var/www/html/`
- Site configs: `/etc/nginx/sites-available/`

Testing Your Installation

1. Create a Test HTML File

Navigate to your NGINX HTML directory and find the `index.html`.

You can also create `nginx_test.html`:

```
<!DOCTYPE html>
<html>
<head>
  <title>NGINX Test</title>
</head>
<body>
  <h1>NGINX is Working!</h1>
  <p>Congratulations! You've successfully installed NGINX.</p>
  <p>Date: <span id="date"></span></p>
  <script>
    document.getElementById('date').textContent = new Date().toLocaleString();
  </script>
</body>
</html>
```

2. Test the File

Visit: <http://localhost> (or :8080 on macOS)

Visit: http://localhost/nginx_test.html (or :8080 on macOS)

```
NGINX is Working!
```

```
Congratulations! You've successfully installed NGINX.
```

```
Date: 9/1/2027, 10:31:18 AM
```

Troubleshooting Common Issues

1. Port Already in Use

Error: `bind() to 0.0.0.0:80 failed`

Solution: Another service is using port 80

```
# Check what's using port 80
netstat -tulpn | grep :80
```

```
# For mac
lsof -i :8080
```

```
# Stop Apache if running (example)
sudo systemctl stop apache2 # Ubuntu
sudo systemctl stop httpd   # CentOS
```

2. Permission Denied (Linux)

Error: Permission denied

Solution: Use sudo or check the firewall

```
sudo nginx  
sudo systemctl start nginx  
  
# Check firewall  
sudo ufw allow 'Nginx Full' # Ubuntu
```

3. NGINX Won't Start (Windows)

Solutions:

- Run Command Prompt as Administrator
- Check if port 80 is free
- Disable Windows World Wide Web Publishing Service:

```
net stop w3svc
```


4. Configuration Errors

Test configuration before starting:

```
nginx -t
```

Common fix: Check syntax in `nginx.conf`

5. Can't Access from Other Devices

Solution: Configure firewall to allow HTTP (port 80)

Success Checklist

NGINX installed using the appropriate method for your OS

NGINX running - can see welcome page

Basic commands work - start/stop/reload

Test file created - custom HTML page loads

Version check - `nginx -v` shows version number

If all are checked, you're ready for the next step.

Configuring NGINX to work with PHP!

Quick Start Summary

Installation Commands

```
# macOS  
brew install nginx && brew services start nginx  
  
# Ubuntu/Debian  
sudo apt update && sudo apt install nginx && sudo systemctl start nginx  
  
# CentOS/RHEL  
sudo yum install nginx && sudo systemctl start nginx
```

Key Takeaways

Why NGINX Matters

1. **Industry Standard:** Used by major websites worldwide
2. **Performance:** Handles many more users than the PHP built-in server
3. **Production Ready:** What you'll use in real applications
4. **Professional Skills:** Essential for web developers

What You've Learned

- How to install NGINX on any operating system
- Basic NGINX commands and file locations
- How to verify your installation
- Common troubleshooting steps